

Why are the bias-corrected forecasts too wide?

Process error, observation error and the one-step-ahead predictive of ORJointUpdate

Claude

October 8, 2026

Summary

The PIT histograms of the bias-corrected runs are centred on 0.5 but humped, so the one-step-ahead predictive is too wide. This note looks for the cause. It reaches four conclusions.

1. **The bias correction does lower the learned process variance, as expected.** By 2017 the median learned process variance v_T is 7–22% of its uncorrected value without the disturbance product (15–58% with it), and it is smaller at 75–94% of sites (Section 2). Without correction the filter raises the process variance after about 1995 to absorb the forecast bias. With correction it keeps falling.
2. **The process variance is no longer the problem; the observation-error variance R is.** After correction, R alone is larger than the mean squared forecast error for three of the four observation-error settings, and close to it for the fourth (Section 3). The correction halves the forecast errors but cannot remove R from the predictive, so the corrected predictive is *more* too wide, relative to its errors, than the uncorrected one ($\text{Var}(z)$ falls from 0.81 to 0.65 for GEDI heteroskedastic R , where $\text{Var}(z) = 1$ means calibrated). A large R also widens the predictive a second way: the analysis barely shrinks the state uncertainty, and that uncertainty is carried into the next forecast.
3. **The observed biomass series are much smoother than R implies** (Section 4). If observation errors were independent with variance R , the year-to-year change Δy_t would have variance at least $R_t + R_{t-1}$. Its robust variance is 12% of that bound for GEDI heteroskedastic R and under 1% for LandTrendr. This holds without running the model. The observation errors must therefore be highly persistent: the implied lag-1 correlation is at least 0.88–0.99. A persistent error adds little to the one-step innovation, but the filter adds the full R to every year’s predictive variance.
4. **The innovations are also heavy-tailed:** most years have very small errors and a few have large jumps, so even a predictive with the right variance would have the wrong shape.

1 Runs, notation and definitions

1.1 Runs

- **Experiments.** The 8 settings of the grid in `OR_joint_update2.Rmd`: observation-error model (GEDI or LandTrendr magnitude $\times$ heteroskedastic or constant R) $\times$ disturbance

product (on or off), all with the gamma prior on τ , at all 538 sites. Timestep $t = 1, \dots, T$, $T = 28$, is calendar year $1989 + t$ (1990–2017).

- **Uncorrected grid:** `experiments/fullRunNoBias`, rerun on October 8, 2026 with the current code and `grid.bias = NULL`. The older `experiments/fullRun` predates the saved process variance, so it is not used here.
- **Bias-corrected grid:** `experiments/fullRunBias`, with `bias = list(rho = 0.9, denom = "N", skip = 1)`.
- **Subset reruns:** `experiments/forecastWidth/{nobias,bias}`. These are 100 randomly chosen sites (`set.seed(1)`) $\times$ 8 experiments $\times$ {uncorrected, corrected}, run again with the same per-run seeds as the grid so that they also store the no-disturbance predictive (Section 1.3). Their learned process variance matches the grid runs exactly (maximum absolute difference 0 over 50 runs checked).

1.2 Model quantities

- y_t : observed above-ground biomass in year t , converted to wood carbon (kg C m^{-2}). x_t : the three VSEM carbon pools; $x_{t,3}$ is wood, the observed pool.
- $R_t = m_R y_t + b_R$: the assumed observation-error variance of y_t ($(\text{kg C m}^{-2})^2$), with $(m_R, b_R) = \text{Rparams}$. Typical values: GEDI heteroskedastic ≈ 0.20 , GEDI constant 0.885, LandTrendr heteroskedastic ≈ 3.1 , LandTrendr constant 3.80. The filter treats the observation errors $\epsilon_t = y_t - x_{t,3}$ as independent across years.
- τ : the precision of the wood-pool process error. On a grid $\tau_1 < \dots < \tau_G$ ($G = 200$, log-spaced over $[0.1, 10^4]$) the filter keeps the posterior $\pi_t(g) = P(\tau = \tau_g \mid y_{1:t})$. $1/\tau_g$ is the process variance added to the wood pool in one year.
- $z_t \in \{0, 1, \dots, D\}$: the disturbance indicator ($0 = \text{undisturbed}$). p_t is its prior, which the disturbance product sets in flagged years.
- Each (grid point g , mixture component $j \in \{1, 2\}$, disturbance class k) has a forecast of y_t 's pool with mean μ_{gjk}^F and wood variance $\Sigma_{gjk,33}^F$. For $k = 0$, $\Sigma_{gj0,33}^F = (\text{importance-weighted ensemble variance of VSEM run from the previous analysis}) + 1/\tau_g$. With bias correction, μ_{gj0}^F includes the shift b_t .

1.3 Plotted and tabulated quantities

Learned process variance $v_t = \mathbb{E}[1/\tau \mid y_{1:t}] = \sum_g \pi_t(g)/\tau_g$ (`hist$procVar`), the posterior mean of the process variance after assimilating y_t . Figure 1 shows its median and interquartile range (IQR) over the 538 sites.

No-disturbance predictive The one-step-ahead predictive of y_t given $y_{1:t-1}$ and $z_t = 0$. It is a Gaussian mixture over (g, j) with weights $\omega_{gj}^0 \propto \pi_{t-1}(g) w_{t-1}(g, j)$ (degenerate components dropped) and components $\mathcal{N}(\mu_{gj0}^F, \Sigma_{gj0,33}^F + R_t)$. Its mean and variance are

$$m_t^0 = \sum_{gj} \omega_{gj}^0 \mu_{gj0}^F, \quad V_t^0 = \sum_{gj} \omega_{gj}^0 (\Sigma_{gj0,33}^F + R_t + (\mu_{gj0}^F)^2) - (m_t^0)^2$$

(new outputs `predMean0`, `predVar0`).

Variance budget $V_t^0 = R_t + Q_t + P_t$, where

- R_t is the observation-error variance;
- $Q_t = \sum_{gj} \omega_{gj}^0 / \tau_g$ (**predProcVar0**) is the process variance expected under the τ posterior π_{t-1} ;
- $P_t = V_t^0 - R_t - Q_t$ is the *propagated state variance*: the uncertainty in the wood pool carried over from the previous analysis through VSEM, plus the spread of the component means μ_{gj0}^F .

Disturbance tail $V_t - V_t^0$, where V_t (**predVar**) is the variance of the full predictive, which includes the disturbed components ($k \geq 1$) with prior weights. This is the extra variance from the low tail of possible disturbances.

Innovation and standardised innovation $e_t = y_t - m_t^0$ and $z_t = e_t / \sqrt{V_t^0}$. For a calibrated Gaussian predictive, $\mathbb{E}[z_t^2] = 1$, the median of $|z_t|$ is 0.674, 95% of $|z_t|$ are below 1.96, and the median of e_t^2 is $0.455 \mathbb{E}[e_t^2]$.

Undisturbed site-years The site-years used in Section 3: $t \geq 3$ (year 1 starts from the population-wide prior on the initial state, and year 2 still carries it), the product flags neither $t - 1$ nor t , and $P(z_t \neq 0 \mid y_{1:t}) < 0.5$. A year counts as flagged if the product's prior $P(z_t \neq 0) > 0.5$.

R scale $c_R = (\bar{e}^2 - \bar{Q} - \bar{P}) / \bar{R}$: the factor on R that would make the mean predictive variance equal the mean squared innovation, keeping Q and P as they are. Bars denote means over undisturbed site-years.

Year-to-year change $\Delta y_t = y_t - y_{t-1}$, over all pairs of consecutive years that the product flags in neither year (all 538 sites, no model involved). The robust variance is $(1.4826 \text{ MAD})^2$, where MAD is the median absolute deviation; it equals the variance for Gaussian data and is not inflated by rare jumps.

2 The learned process variance

experiment	median v_t , 1995		median v_T , 2017		median $v_T^{\text{corr}} / v_T^{\text{uncorr}}$	sites with smaller v_T
	uncorr.	corr.	uncorr.	corr.		
GEDI het., no product	0.106	0.031	0.207	0.0119	0.08	94%
GEDI const., no product	0.103	0.076	0.252	0.0132	0.07	92%
LandTrendr het., no product	0.196	0.184	0.165	0.0244	0.18	87%
LandTrendr const., no product	0.216	0.208	0.150	0.0280	0.22	86%
GEDI het., product	0.082	0.037	0.140	0.0091	0.15	92%
GEDI const., product	0.112	0.082	0.113	0.0170	0.22	86%
LandTrendr het., product	0.209	0.206	0.081	0.0360	0.58	75%
LandTrendr const., product	0.226	0.232	0.084	0.0404	0.57	75%

Table 1: Learned process variance v_t ($(\text{kg C m}^{-2})^2$) in 1995 and 2017, and the per-site ratio of corrected to uncorrected v_T .

You expected the process variance learned with bias correction to be smaller. It is, by a lot (Table 1).

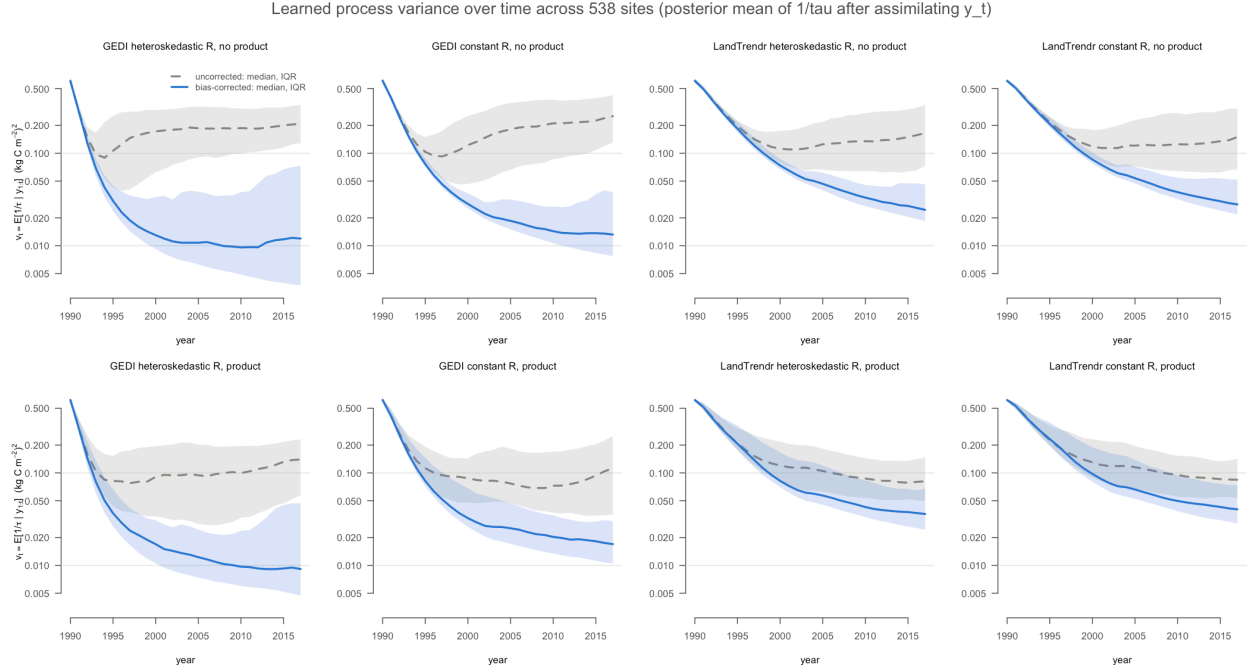


Figure 1: Learned process variance $v_t = \mathbb{E}[1/\tau \mid y_{1:t}]$ over time (log scale). Lines are the median over 538 sites and bands the IQR. Gray dashed: uncorrected; blue: bias-corrected. Rows: without and with the disturbance product.

- **Without correction, the process variance absorbs the bias.** In the GEDI experiments without the product (and, more weakly, GEDI heteroskedastic R with it) the uncorrected v_t falls until about 1995 and then *rises*, to about 0.2 by 2017. The forecast errors there have a consistent sign (VSEM over-predicts wood). The only way the uncorrected filter can account for errors that large is a larger process variance, so the τ posterior moves to smaller τ . With correction, the systematic part is removed and v_t keeps falling, to about 0.01, a process standard deviation of about $0.1 \text{ kg C m}^{-2} \text{ yr}^{-1}$.
- **With LandTrendr error, the two curves agree until about 2000.** With $R \approx 3\text{--}4$ each observation says little about τ , so v_t falls slowly and the bias only starts to matter after several years of data. The corrected curve still ends 5–7 times lower (without the product).
- **The product shrinks the effect.** In flagged years the predictive includes the disturbance components, and the filter can explain a low observation as a disturbance instead of through τ . The uncorrected v_t is therefore smaller with the product, and the gap to the corrected run is smaller too.
- **The τ grid limit is not binding.** The smallest process variance on the grid is $1/\tau_{\max} = 10^{-4}$, and no site has $v_T < 10^{-3}$ in any experiment. The posterior is not stuck at the edge of the grid.

So the process variance is not why the corrected forecast is too wide: it is already small and became smaller with correction.

3 Variance budget of the predictive

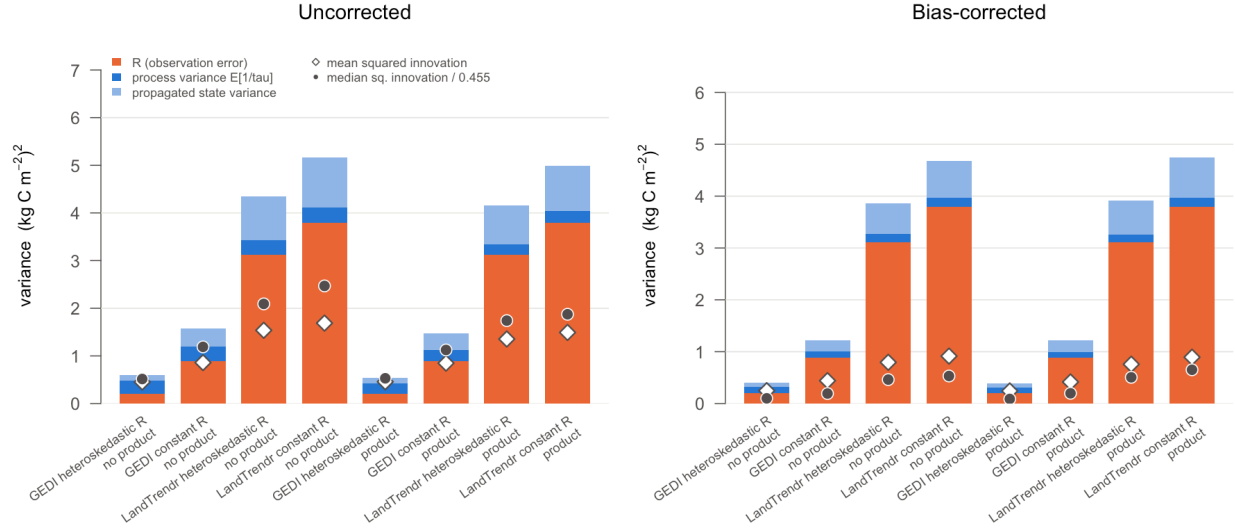


Figure 2: Mean variance budget $\bar{V}^0 = \bar{R} + \bar{Q} + \bar{P}$ of the no-disturbance predictive over undisturbed site-years (100-site subset), against the mean squared innovation $\bar{e}^2$ (diamonds) and the median squared innovation divided by 0.455 (dots; the median of e^2 is $0.455 \text{ Var}(e)$ for Gaussian e). The predictive is calibrated when the bar height equals the markers.

experiment (no product)	corr.	$\bar{e}^2$	med e^2	$\bar{R}$	$\bar{Q}$	$\bar{P}$	$\bar{V}^0$	$\mathbb{E}[z^2]$	med $ z $	c_R
GEDI het.	no	0.452	0.235	0.201	0.284	0.115	0.600	0.81	0.68	0.27
	yes	0.248	0.046	0.201	0.124	0.072	0.398	0.65	0.39	0.26
GEDI const.	no	0.860	0.541	0.885	0.318	0.368	1.570	0.57	0.60	0.20
	yes	0.441	0.088	0.885	0.124	0.213	1.220	0.35	0.28	0.12
LandTrendr het.	no	1.540	0.952	3.120	0.312	0.914	4.340	0.35	0.48	0.10
	yes	0.796	0.210	3.120	0.150	0.598	3.860	0.20	0.23	0.02
LandTrendr const.	no	1.690	1.120	3.800	0.314	1.050	5.160	0.32	0.47	0.09
	yes	0.915	0.242	3.800	0.164	0.708	4.670	0.19	0.23	0.01

Table 2: Variance budget of the no-disturbance predictive over undisturbed site-years (about 1840 per row), in $(\text{kg C m}^{-2})^2$. Calibrated values: $\mathbb{E}[z^2] = 1$, median $|z| = 0.674$, $c_R = 1$. The runs with the product give nearly the same numbers; for LandTrendr, c_R is slightly negative with the product (the Q and P parts alone exceed $\bar{e}^2$).

Table 2 and Figures 2–3 show where the predictive width comes from.

- **The correction reduces the errors more than the width.** For GEDI heteroskedastic R , the mean squared innovation halves (0.45 to 0.25) and the process part Q falls from 0.28 to 0.12. But $R = 0.20$ stays in the predictive regardless, so V^0 falls only from 0.60 to 0.40. As a result $\mathbb{E}[z^2]$ drops from 0.81 to 0.65: relative to its errors, the corrected predictive is more too wide than the uncorrected one. The same holds in every experiment. This is why the corrected PIT histograms are more humped than you might expect. Before correction, the bias inflated the errors enough to roughly match the wide predictive.

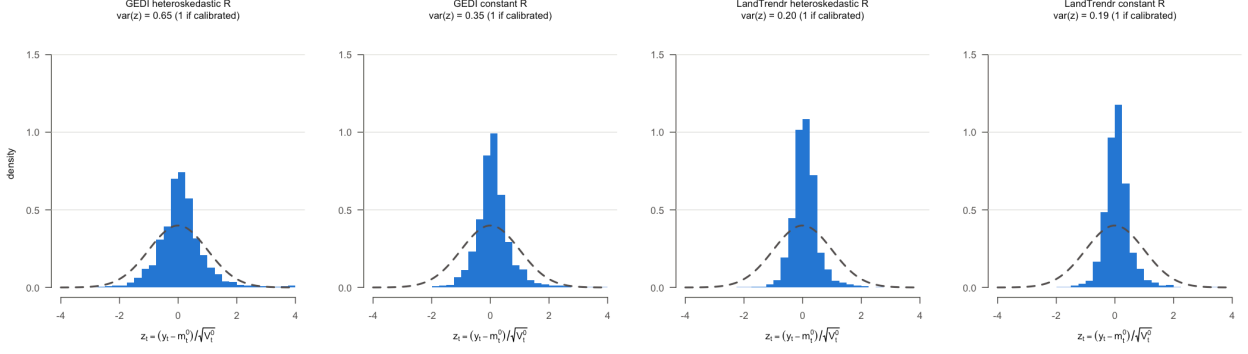


Figure 3: Standardised innovations $z_t = (y_t - m_t^0) / \sqrt{V_t^0}$ of the bias-corrected runs without the product, over undisturbed site-years. Dashed: $\mathcal{N}(0, 1)$, the calibrated reference.

- **R alone is too large.** For GEDI constant R and both LandTrendr settings, $\bar{R}$ is larger than the whole mean squared innovation (0.885 vs 0.44, 3.12 vs 0.80, 3.80 vs 0.92). Even with zero process variance and perfect knowledge of the state, the predictive would be too wide. To match the mean squared innovation, R would have to be scaled by $c_R \approx 0.26$ (GEDI heteroskedastic), 0.12 (GEDI constant) and 0.01–0.02 (LandTrendr).
- **R also widens the predictive through P .** For a scalar Kalman update, the analysis variance is $P^a = P^f R / (P^f + R)$. When $R \gg P^f$, $P^a \approx P^f$: the observation barely reduces the state uncertainty, and that uncertainty is carried into the next forecast. The propagated part is $\bar{P} \approx 0.07$ for GEDI heteroskedastic and 0.6–0.7 for LandTrendr. So a large R widens the predictive twice: directly, and through a poorly constrained state.
- **The innovations are heavy-tailed.** For a Gaussian, the median of e^2 is $0.455 \mathbb{E}[e^2]$. For the corrected runs it is 0.18–0.26 $\mathbb{E}[e^2]$: most years have very small errors and a few have large ones. Even for GEDI heteroskedastic, where the average variance is closest to calibrated ($\mathbb{E}[z^2] = 0.65$), the typical $|z|$ is 0.39 against 0.674, and 97% of $|z|$ fall within ± 1.96 (expected 95%). A Gaussian predictive with the right variance is still too wide in typical years and too narrow in jump years (Figure 3).
- **The disturbance tail adds width but not the hump.** The disturbed components add about 1.1–1.25 to the full predictive variance ($V_t - V_t^0$) in every experiment. That variance sits in a low tail holding only the background disturbance probability (about 5% per year). It moves the PIT of an undisturbed year by about that 5%, but it is not what makes the hump, which already appears in z_t (Figure 3).

4 Why R is too large: the observations are smooth in time

This section uses only the data, not the filter. Suppose $y_t = x_{t,3} + \epsilon_t$ with independent ϵ_t of variance R_t . Then

$$\text{Var}(\Delta y_t) = \text{Var}(\Delta x_{t,3}) + R_t + R_{t-1} \geq R_t + R_{t-1}.$$

Now allow the errors to be correlated, with lag-1 correlation ρ and equal variance R . Then $\text{Var}(\Delta \epsilon_t) = 2R(1 - \rho)$, so $\rho \geq \rho_{\min} = 1 - \text{Var}(\Delta y) / (2\bar{R})$.

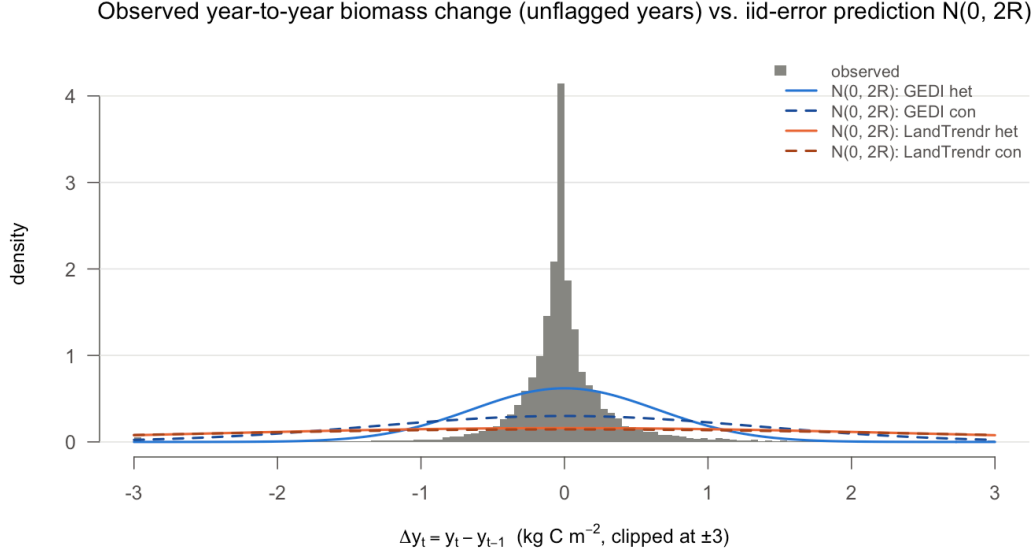


Figure 4: Histogram of the observed year-to-year change Δy_t over 10 720 pairs of consecutive years not flagged by the product (538 sites), against $\mathcal{N}(0, 2\bar{R})$, the smallest spread that independent observation errors with variance R allow, for each observation-error model.

observation-error model	$2\bar{R}$	$\text{Var}(\Delta y)/2\bar{R}$	robust $\text{Var}(\Delta y)/2\bar{R}$	$\rho_{\min}$
GEDI heteroskedastic	0.414	0.70	0.117	0.88
GEDI constant	1.770	0.16	0.027	0.97
LandTrendr heteroskedastic	6.240	0.046	0.0077	0.99
LandTrendr constant	7.600	0.038	0.0064	0.99

Table 3: Observed year-to-year change against the lower bound $2\bar{R}$ implied by independent observation errors. $\text{Var}(\Delta y) = 0.290$ and robust $\text{Var}(\Delta y) = 0.048$ (kg C m^{-2})², the same for every row. $\rho_{\min}$ uses the robust variance.

- **The observed changes are far smaller than independent errors allow.** The robust variance of Δy is 12% of $2\bar{R}$ for GEDI heteroskedastic and under 1% for LandTrendr (Table 3, Figure 4). The median $|\Delta y|$ is 0.10 kg C m^{-2} .
- **The series look smoothed.** 21% of year-to-year changes are exactly zero, and the values come on a 0.05 kg C m^{-2} grid (1 Mg ha^{-1} of biomass). The changes are *positively* correlated from one year to the next (lag-1 correlation $+0.32$). Independent errors would make it negative, since $\text{Corr}(\Delta \epsilon_t, \Delta \epsilon_{t-1}) = -1/2$. This is what a temporally fitted product looks like (for example LandTrendr’s piecewise-linear segmentation): the trajectory is smooth, and its error relative to the truth is persistent from year to year.
- **This explains the size of c_R .** With persistent errors, a site that is over-estimated one year is over-estimated the next. In the one-step innovation, most of ϵ_t is already explained by ϵ_{t-1} , which the previous analysis partly absorbed into the state. For a first-order autoregressive (AR(1)) error, the part of ϵ_t that ϵ_{t-1} cannot predict has variance $R(1 - \rho^2)$. At $\rho_{\min}$ this is $0.22R$ (GEDI heteroskedastic), $0.05R$ (GEDI constant) and $0.01\text{--}0.02R$ (LandTrendr), close to the scales c_R the innovations ask for in Table 2 (0.26, 0.12, 0.01–0.02). The filter instead

adds the full R to every year’s predictive variance.

- **Why the stated R is not wrong as such.** The `Rparams` describe the error of a single biomass map relative to the truth, which is the right quantity for one snapshot. It is the wrong quantity for year-to-year innovations when that error persists over time.
- **The raw variance hides this.** The non-robust ratio for GEDI heteroskedastic is 0.70, much closer to 1, because a few large jumps (unflagged disturbances and product artefacts) dominate the plain variance. This is the same heavy tail seen in the innovations.

One consequence for the bias correction: b_t estimates the persistent offset between the observations and the undisturbed forecast. At a given site that offset is the VSEM bias *plus* the site’s persistent observation error, and the filter cannot separate the two. The correction therefore absorbs part of the persistent observation error, which helps the innovations.

5 Implications

1. **Use an innovation-scale observation error.** Replace R in the filter with the variance of the non-persistent part, $R(1 - \rho^2)$, with ρ estimated per product, or estimate a scale c on R from the data. A scale can go on a grid jointly with τ , using the same machinery as the τ posterior. The innovations point to $c \approx 0.1$ – 0.3 for GEDI and about 0.01 – 0.02 for LandTrendr.
2. **Or model the persistence explicitly.** Add the observation-error offset to the state as an AR(1) term. The bias b_t already works like a slowly updated version of this offset; modelling it would let the filter separate VSEM bias from observation offset over many sites.
3. **Then handle the heavy tails.** After the scale is fixed, a Student- t or two-component observation or process error would cover the rare jumps without widening every year.
4. **Re-check the disturbance product’s flags.** The second PIT peak near 0.8 (not analysed here) comes from flagged years without a visible biomass drop.

6 Reproducing

- Code changes: `ORJointUpdate` now also returns `predMean0`, `predVar0` and `predProcVar0`, the no-disturbance predictive defined in Section 1.3.
- Uncorrected grid: `Rscript run_grid.R` with `grid.outdir <- "experiments/fullRunNoBias"` and `grid.bias <- NULL` in the `run-grid` chunk.
- Figures and every number in this note: `Rscript claude-working-docs/forecast_width_analysis.R`, from the repository root (set `EXP_DIR` if the experiment folders are elsewhere). The script reruns the 100-site subset into `experiments/forecastWidth` if it is missing.